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Web Technology

Quality Laboratory Manual

Experiment No. 04

Program Based on JQuery selectors and JQuery events



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Experiment No. 04

Title of Experiment: Program Based on JQuery selectors and JQuery events

Aim of Experiment: To develop a dynamic web page using jQuery selectors and events to manipulate HTML elements and handle user interactions.

System Requirements –

Operating System: Windows 10 or above

RAM: Minimum 4 GB

Software/s Needed for Experiment – Web Browser, Text Editor, VSCode

Experiment Objectives:

- To understand the fundamentals and usage of the jQuery library in web development.
- To learn different types of jQuery selectors for accessing HTML elements.
- To implement jQuery events for handling user interactions.
- To manipulate HTML content dynamically using jQuery and DOM.
- To develop interactive and responsive client-side web pages using jQuery.

Experiment Outcomes:

- Understand the basic concepts and advantages of jQuery in client-side scripting.
- Apply various jQuery selectors to select and manipulate HTML elements.
- Implement jQuery event handling for dynamic user interaction.
- Modify web page content dynamically using jQuery DOM manipulation.
- Design interactive and user-friendly web applications using jQuery.

Theory:

jQuery is a lightweight JavaScript library used for DOM manipulation, event handling, animations, and AJAX interactions. JQuery selectors help in selecting HTML elements using id, class, or tag names. JQuery events allow execution of functions when users interact with web elements such as clicking buttons or submitting forms.

1) JQuery Selectors

jQuery selectors are used to select and access HTML elements so that developers can manipulate their content, style, or behavior easily. They work similar to CSS selectors and allow targeting elements by id, class, tag name, or attributes. Once selected, actions like hiding, showing, or modifying elements can be performed using jQuery methods.

Common Types of jQuery Selectors:

- **ID Selector:** selects a single element using id → `$("#id")`
- **Class Selector:** selects elements using class name → `$(".class")`
- **Element Selector:** selects all elements of a specific tag → `$("p")`
- **Universal Selector:** selects all elements → `$("*")`

- **Attribute Selector:** selects elements based on attributes → `$ (" [ type = ' text ' ] ")`

2) JQuery Events

JQuery events are used to perform actions when users interact with web page elements such as clicking buttons, entering data, or submitting forms. They make web pages dynamic and responsive by allowing developers to attach functions that execute automatically when an event occurs. jQuery simplifies event handling with easy and readable syntax compared to traditional JavaScript.

Common jQuery Events:

- **click()** – triggered when an element is clicked
- **change()** – triggered when input value changes
- **submit()** – triggered when a form is submitted
- **hover()** – triggered when mouse moves over an element
- **dblclick()** – triggered on double click

Program Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>jQuery Selectors and Events Example</title>

  <!-- jQuery CDN -->
  <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>

  <style>
    body{
      font-family: Arial;
      padding:20px;
    }
    .box{
      width:200px;
      height:100px;
      background-color:lightblue;
      text-align:center;
      line-height:100px;
      margin-top:10px;
    }
    .highlight{
      background-color:yellow;
    }
  </style>
</head>
<body>
  <h2>jQuery Selectors and Events Demo</h2>
  <p id="para">Click the button to change my color.</p>
  <input type="text" id="textbox" placeholder="Type something...">
  <p class="output"></p>
  <button id="btn">Click Me</button>
```

```
<div class="box">Hover on Me</div>
<script>
$(document).ready(function(){

    // ID Selector with Click Event
    $("#btn").click(function(){
        $("#para").css("color","red");
    });
    // Class Selector with Hover Event
    $(".box").hover(function(){
        $(this).toggleClass("highlight");
    });
    // Keyup Event with Element Selector
    $("#textbox").keyup(function(){
        $(".output").text("You typed: " + $(this).val());
    });
});
</script>
</body>
</html>
```

Task for Students: Design “Mouse Hover Animation” using jQuery.

Task Requirements:

- Create an HTML web page and include **jQuery CDN**.
- Use **jQuery selectors** to select elements (ID/Class/Element)
- Apply **hover() event** to perform at least two animation effects (fade, slide, size, color change).
- Design a simple UI using CSS and ensure animation works on mouse enter and leave.

Submission Requirement:

- Submit the **HTML file** with jQuery code (and CSS file if separate).
- Attach **output screenshot** showing hover animation.

Observations:

The jQuery hover event successfully detected mouse enter and leave actions on the selected element. Animation effects like color change and size variation were applied smoothly. Selectors helped in targeting specific elements easily without complex code. The web page became more interactive and responsive due to jQuery events.

Conclusion:

This experiment demonstrated how jQuery selectors help in easily selecting and manipulating HTML elements. Students learned to handle user interactions using different jQuery events such as click, hover, and keyup. The program showed how dynamic changes and animations can be implemented with simple and minimal code. Overall, the experiment improved understanding of event handling, DOM manipulation, and creating interactive web pages using jQuery.

Expected Oral Questions:

1. What is jQuery and why is it used in web development?
2. What are jQuery selectors? Give examples.
3. Explain ID selector and Class selector in jQuery.
4. What is an event in jQuery?

FAQs in Interview:

1. What is jQuery?
2. What are jQuery selectors and why are they used?
3. Explain different types of selectors in jQuery.
4. What is the difference between ID and Class selector?
5. What are jQuery events? Give examples.